

Challenges and Opportunities in the Responsible Use of AI and Human-Computer Interaction

Laura Steckman

Air Force Research Laboratory
Arlington, Virginia, USA
laura.steckman.1@us.af.mil

Nandini Iyer

Air Force Research Laboratory, Air
Force Office of Scientific
Research/European Office of
Aerospace Research and Development
London, United Kingdom
nandini.iyer.2@us.af.mil

Boyoung Kim

George Mason University Korea
Incheon
bkim55@gmu.edu

Joseph Lyons

Air Force Research Laboratory
Wright-Patterson AFB, Ohio, USA
joseph.lyons.6@us.af.mil

Massimiliano Cappuccio

School of Engineering and
Information Technology
University of New South Wales
Canberra, NSW, Australia
m.cappuccio@unsw.edu.au

Mark Richard Gaffley

Private Law
University of Cape Town
Cape Town, Western Province, South
Africa
gffmar002@myuct.ac.za

Elizabeth Phillips

George Mason University
Fairfax, Virginia, USA
ephill3@gmu.edu

Sonia Sousa

School of Digital Technologies
Tallinn university
Tallinn, Estonia
scs@tlu.ee

Max Van Kleek

Department of Computer Science
University of Oxford
Oxford, Oxfordshire, United Kingdom
max.van.kleek@cs.ox.ac.uk

Abstract

The use of Artificial Intelligence (AI) is fraught with ethical challenges as we increasingly start to use and rely on AI in society. Despite the potential promises of AI to enhance human abilities and increase how Human-Computer Interaction (HCI) utilizes HUMAN-centered inputs, issues arise wherein AI or its outputs violate human ethics and/or social norms. In response, the academic, industrial, and military research communities have advocated for increased research to understand how to best develop, test, and use AI technologies and their outputs in ways that are consistent with our ideals, values and morals. Fields of inquiry, such as machine ethics, have burgeoned as a result. Yet, moral and ethical challenges associated with AI and HCI persist. Recent literature on AI ethics and responsible use of AI remains limited to a small number of principles. HCI requires considering a larger number of principles for human-centered design, or even the incorporation of ethics and values more holistically, yet remains limited without a solid, more inclusive scientific foundation. The current panel will bring together diverse perspectives in discussing issues associated with research related to the responsible use of AI and HCI. The participants, along with the expert panelists, will play a key role in shaping the direction of the conversation through active participation.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Philosophical/theoretical foundations of artificial intelligence; • **Human-centered computing** → Human computer interaction (HCI); HCI theory, concepts and models; Human computer interaction (HCI); HCI design and evaluation methods; • **Social and professional topics** → Professional topics; Computing profession; Codes of ethics; • **General and reference** → Cross-computing tools and techniques; Design.

Keywords

Value-based design, Human-Centered design, Value alignment

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1 INTRODUCTION

The development and use of artificial intelligence (AI) in society has the potential to dramatically shape society. This shaping is particularly revealing within the context of HCI, where AI has allowed emerging technologies and approaches to consider nuanced human data in new, improved ways (i.e. human-centered design) and moved toward interactions with AI and AI-enabled systems. Human interactions with computer-based technologies are increasingly stretching the boundaries of human comfort and preference.

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AI, especially in the context of HCI where people will interact with self-adapting and semi- to fully-autonomous systems, is believed to have a significant role in shaping the future of education, health-care, transportation, and the military [1–3]. AI has the potential for massive computational power, omnipresent reach, and ubiquitous presence; in contrast, humans have known constraints and limitations on their ability to process sensory and perceptual information as well as biases. However, humans, also possess innate and learned social awareness, contextual awareness, and can be adept at making decisions under complexity. In fact, Klein [4] exemplifies the swiftness of human pattern recognition in dynamic situations as humans make decisions in dangerous and otherwise complex situations. The recent problems confronting societies however seems to have outpaced human abilities to find timely, effective solutions. Therefore, society has an obligation to shape how AI and AI for HCI are developed to ensure that they are used in ways that take advantage of the strengths, but also reflect our societal values and in ways that advance human and human-computer moral behavior. Similar goals have been noted within the Department of Defense (DoD) as the DoD seeks to ensure that novel technology is used in ways that are consistent with human values [5, 6]. The stakes couldn't be higher and the timing any better as multiple national and multinational meetings and strategies have focused on responsible use of AI (RAI) over the past two years. The European Union's AI Act was a world first in the regulation of AI. In November 2023, the Presidents of the United States of America and China met to deliberate and agree upon some guidelines for the regulation of AI in defense systems. This meeting followed President Biden's Executive Order on Trustworthy, Safe, and Secure AI, which the United Nations adapted and adopted in early 2024. The world, including the DoD, may benefit from AI, however, the use of AI needs to be consistent with human value structures. Policies, processes, and guardrails need to be put in place to ensure that our adoption of AI in various forms within society is done responsibly.

After AI gained additional computational power and the span of delegated authority increased to bring the need for RAI more prominently to the forefront, HCI underwent a similar shift as the field started to move away from technology-centered design to human-centered design with a need to consider values and norms. Humans are increasingly looking to ethical theories as a framework for considering human-centered design and the responsible use of AI [7]. Ethics can be considered as the combined (and accepted) norms, rules, precepts, and guiding principles that influence the attitudes and behavior of humans [8]. Ethics in this context take on the form of norms within a social group. Normative approaches to ethical theories often involve virtue ethics (understanding the characteristics and features of ethical agents [i.e., is this person good or bad, is this person competent – incompetent]; deontological ethics which emphasizes the rule-based nature of behaviors (i.e., is this behavior good or bad; is this behavior consistent or divergent from a rule); and considerations of the behavioral outcome (i.e., is this a good or bad outcome independent of what was intended) [8]. Based on these, and other ethics theories that may prove relevant to intelligent technologies, there is a growing need to build frameworks, methods, approaches, and measurements for embedding ethics appropriately into systems to develop AI and HCI responsibly. The need increases as technology continues to develop

faster and faster, as demonstrated by the increased conversations—and controversy—driven by public access to and subsequent media fervor over Large Language Models (LLMs), such as ChatGPT, over the past two years. In many ways, technology development has outpaced the ability of society to evaluate the appropriateness of said technology.

Technology development and deployment incorporates value systems; with the current rate of technology growth, these invisible value systems affect and influence users and consumers continuously across the technological lifecycle. Value embeddedness affects people in various ways, both foreseen and unforeseen, based on whose values are inserted at which point in the process, and how they inform the decisions or conclusions the systems draw. Despite understanding values within technologies create effects, AI ethics is a relatively new field. RAI is even newer. Their growth follows that of the movement toward “trustworthy AI,” but has often been confused with facets and components of that concept. For example, ethics, which can refer to non-maleficence, is often confused with assurance, ensuring the system works as intended at an acceptable risk level, or lumped together with it under the umbrella of safety. This early science affects not only AI Ethics and RAI research, but also spills over to the HCI community as it incorporates AI, ML, and grapples with questions about values and ethics. This example exemplifies the importance of establishing some common ground within the research community.

Because of the complexity involved in ethics and technology, the nature of which requires the multidisciplinary collaboration between often non-convergent scientific disciplines, some researchers believe it is a fruitless endeavor [9]. However, such sentiment remains low, as demonstrated by the Artificial Intelligence Index Report 2023, which devotes an entire chapter to how researchers and practitioners continue to focus on and grow AI ethics [10]. The ChatGPT example above is a primary exemplar; despite LLMs having existed for more than three decades, the public's access to the technology through beta testing has only now launched an entire line of research on the ethicality and bias associated with system usage [11–16], which even includes research about whether ChatGPT should be biased [17]. Such developments, with the technology and the research, suggest an even greater need for AI ethics research at faster speeds and greater scales. This is also why the authors believe that a panel focused on these issues is central to communities such as CHI, among others.

There are two primary areas where AI ethics and HCI research gaps create societal tensions. The first is that, despite acknowledgement of the importance of ethics, the research itself has focused on critical topics, such as fairness, bias, and transparency [18], that address pieces of value systems but have yet to focus on the larger picture. The aforementioned 2023 AI report limits its discussion of AI ethics solely to fairness and bias, showcasing current limitations of the research. In contrast, multiple meta-analyses of the global state of AI ethics and policy research found more than 20 shared principles that constitute AI ethics [18, 19]. These additional principles have not yet entered mainstream AI ethics research, which leaves the HCI community to struggle with less RAI knowledge than it requires, among other challenges [20]. The second challenge for AI ethics' research is a theory-practice gap between ethics and engineering. While ethics engineering is often part of formal education,

it has often been left in the classroom. While some approaches have been developed to address the theory-practice gap, such as IEEE's Ethically Aligned Design guidebook [21] and an embedded ethics approach to engineering [22], more research is required to advance the field so that it can fuse ethics with engineering [23]. The latter would greatly benefit HCI and human-centered design research. Moving ethics research into engineering and design practice is one of the next biggest challenges to meet national and international requirements.

Recognizing not only the need for the research, as well as the extent to which AI ethics might embrace and extend beyond its current limitations, the US Air Force Research Lab facilitated an international, multidisciplinary workshop to determine current AI ethics research gaps and needs in early 2023 [24]. The participants, representing disciplines from philosophy and social science to robotics and computer science, developed nine recommendations to advance the science surrounding AI ethics and human-technology interactions with HCI relevance. Beyond those recommendations, the workshop demonstrated the need to have a more sustained conversation with an even more diverse audience. To that end, the proposed panel brings together a diverse, qualified panel to share their expertise and engage participants in the start to that wider conversation that starts applying AI and ethics so they, together, can transition to responsible AI, the apparent emerging next phase of AI and HCI development.

There are three additional reasons why the HCI community should care about the science of AI ethics and these are driven by empirical findings that characterize asymmetries that exist between humans and AI. First, humans tend to have biased preference toward humans over AI during ethical decision making primarily through algorithm aversion [25]. The HCI community will need to grapple with and understand these preferences should it seek to use AI in complex ethical scenarios. Second, humans tend to react with less intensity to ethical violations from AI relative to violations from humans. In a process referred to as algorithmic outrage deficit, Bigman and colleagues [26] found that humans report lower moral outrage for discrimination across a range of contexts. This effect is driven by lower perceived intent to discriminate, and interestingly, this effect appears to hold up for human-AI mediated contexts as well [27]. Finally, there are growing concerns in the HCI community regarding the capability of AI to motivate unethical behaviors by humans [28]. Thus, the HCI community is faced with an overall preference for humans over AI decision, a concomitant enervated reaction to AI during ethical violations. These issues could create cascading negative effects within the HCI if left unaddressed.

1.1 Applying ethical theories to AI and HCI

There are multiple issues in applying normative ethical theories to AI technologies. This is particularly challenging for applying normative theories of ethics to machines because humans have difficulties in viewing machine as moral entities [29]. These issues are elaborated in [30, 31] and involve challenges in the application of virtues to machines, lack of prescribed intent to machine behaviors, asymmetries in how machines are viewed relative to humans, and preferences for humans over machines in complex situations. Virtues such as competence and benevolence are critical

factors in shaping perceptions of ethicality. Competence is a critical virtue as it relates to one's ability to perform in specific situations. Benevolence is also a key aspect of ethical behavior as it relates to one's intentional stance and goals – and in particular perceived benevolence will shape one's attitude toward a benevolent referent. One of research challenges associated with the use of virtues to describe machines include: 1) it is unclear what virtues make sense in the context of AI-based technologies; and 2) virtues as ascribed to humans likely do not convey the same impact when applied to machines (consider the asymmetries of perceived competence comparing humans to machines due to processes such as algorithm aversion), and finally attributions related to intent (e.g., benevolence) are difficult to understand in the context of machines – whereas these are intuitive attributions for human referents. Even when machines are objectively better than humans at particular tasks, there are still situations wherein humans will prefer other humans. In prior research on trust dynamics, researchers identified clusters of humans in terms of their trust dynamics finding that some humans (characterized as disbelievers) will never establish trust of a technology despite evidenced reliability [28]. While the literature in this domain is burgeoning, a couple notable studies have found that the moral behaviors of robots are not viewed the same ways as those of humans. Research by [32] found that robots are punished for making utilitarian decisions that benefit those who caused the situation (drunk boaters causing an accident for example) whereas humans are not punished for such decisions. Further, research by [33] found that robot nurses were perceived negatively when they forcibly applied medication, but no such penalties were applied to human nurses.

2 RESPONSIBLE USE OF AI

There has been a strident increase in demand on responsible use of AI, yet, there remains a need to develop a scientific foundation for measuring and meeting standards of responsibility. More recent research [1] has discussed five areas wherein researchers and practitioners could advance responsible use of AI 1) education in ethics and core AI concepts for AI developers, leaders, and users; 2) development and use of model cards or datasheets for datasets to provide transparency into the strengths, limits, and potential biases of a trained model; 3) employing human-centered design that seeks to understand human value structures within a task context and enable effective human-machine interaction through intuitive and transparent interfaces; 4) targeted use of run time assurance that monitors and modifies the inputs or outputs of a trained model when necessary to enforce ethical principles such as safety or limiting bias; and 5) developing best practices for the use of a joint human-AI co-creation and training experience to enable a shared mental model and higher performance through potential emergent behavior. There is a need for additional research to forge a deeper understanding of how to develop, test, and use AI responsibly

3 PANEL RATIONALE

The panelists were selected based on a motivation for technical expertise, diversity in geographic region (hence perhaps diversity of value structures), and diversity in academic tenure. The moderators sought to represent perspectives from graduate students, junior

faculty, and more experienced perspectives. The organizers had a pre-existing relationship with some, but not all, of the panelists to ensure that multiple, diverse perspectives would be represented. The panel consists of the following:

MASSIMILIANO L. CAPPUCIO is Deputy director of the Values in Defense and Security Technology group at the University of New South Wales, Canberra, Australia. His research in human performance and human-machine interaction attempts to combine different embodied approaches to cognition. As a cognitive philosopher and a technology ethicist, his research on intelligent systems is interdisciplinary and aims to integrate phenomenological analyses, empirical experimentation, and synthetic modeling. His current research focuses on skill acquisition and disruption, social-robotics, theory and ethics of artificial intelligence and autonomous systems. He has also served as organizer of academic events, including interdisciplinary workshops, research seminar series, and international conferences (like the TEMPER workshop on Training, Enhancement, and Military Performance and the annual Joint UAE Symposium on Social Robotics).

MARK GAFFLEY from Cape Town, South Africa, is an admitted attorney with over twelve years of experience in start-up, in-house legal, and research roles. He has worked with leading local and international organizations, including Uber, Takealot, Media24, the Human Sciences Research Council, simple.Capital(), and the Global Center on AI Governance. From 2019 to 2024, he ran his own legal and research consultancy, focusing on law-centered artificial intelligence research and advisory projects for several of these organizations. Mark holds degrees in Arts and Law from the University of Cape Town (UCT) and is in the final stages of completing a PhD in Jurisprudence at UCT. His thesis critically examines the confrontation between algorithmic norms and the Kantian ethical orientation, the foundational framework of modern constitutional systems, with a particular focus on the impact of artificial intelligence on human autonomy, independent decision-making, and critical thinking.

ELIZABETH “BETH” PHILLIPS is an Assistant Professor and junior faculty member in the Human Factors and Applied Cognition Group in the Department of Psychology at George Mason University. She is also the director of the Applied Psychology and Autonomous Systems (ALPHAs) Laboratory. Broadly, her work focuses on interactions with robots, autonomous systems, and related technologies like virtual and augmented reality. Her recent work investigates the role of justification of robots’ moral decision making as well as the application of role ethics for language capable robots, and how people perceive robots as moral agents. As a woman roboticist, Dr. Phillips is also committed to increasing access to tech space for historically underrepresented groups. She has served as an organizer and fundraiser for the Meet the Women in Robotics Workshop series at the Robotics: Science and Systems Conference, as a mentor for the Feminist Leadership and Mentorship for Equality Workshop, and as a mentor of students in her graduate university’s Direct Connect program which transitions students from community college to a four-year university.

SONIA CLAUDIA DA COSTA SOUSA is Associate Professor of Interaction Design at the School of Digital Technologies, Tallinn University, Estonia. She holds two PhDs (2006 & 2023) and post-doc projects (2010 & 2013). Her research interests are focused on

Trust in technology, Trustworthy AI, Human-Computer Interaction, and User experience. She has been researching and mapping the Trust implications in Human-Computer Interaction (HCI) for over 15 years. She is the Estonian Tallinn ACM-W Chapter chair member and vice chair of the Estonian ACM SIGCHI Chapter. In the past (2019) she was the chair Mozambique ACM-W Chapter representative. She has contributed to organizing scientific dissemination events like INTERACT 2019, NordiCHI 2020, AfriCHI 2021, ICIDS 2021 & 2022, and The European Conference on Cognitive Ergonomics 2025. Her most recently funded projects include NGI-Trust, CHIST-ERA, Horizon 2020, and AFOSR (U.S. Air Force Office of Scientific Research).

MAX VAN KLEEK is Associate Professor in the Department of Computer Science at the University of Oxford, where he co-leads the Human Centered AI (HCAI) sub-theme, and LGBTQ+ Fellow of Kellogg College, Oxford. His research focuses on the design of ethical and inclusive systems, that centre on privacy, empowerment and respect. He has over 20 years of research experience within Human-Computer Interaction and AI, including at MIT CSAIL, MIT Media Lab, (Xerox) PARC, IBM Research, and Nokia, more than 150 peer reviewed papers, and led 2 UKRI research grants under the £13.8m PETRAS Programme (Privacy Ethics Trust Reliability Accountability and Security). He has recently become the co-founder of an Oxford AI spin-out: Veer Technologies.

4 PANEL FORMAT

The panel is intended to maximize participation from the audience and will be highly interactive and energetic. Upon entrance to the panel session, participants will scan the QR code for the active Slido application for this panel using their mobile device. The code will connect each participant to the panel session. Once participants are connected to the Slido, the session will start with a pre-session poll asking participants which ethical or RAI principle(s) HCI research should focus on next. At the session start, the moderators will introduce themselves and the panelists. Next, the panelists will offer a brief (e.g., 5-min) teaser statement related to responsible use of AI and HCI linking to a prior or ongoing research project. Following the opening remarks, the moderators will display three questions and the audience will select which one to answer first via the Slido app (as a warm up exercise). The moderators will display the results in real-time and each panelist will make an attempt to answer that question. Example questions are below:

- How do you use/view responsible AI in your work?
- What are the biggest barriers to society when using AI in HCI?
- What has been the biggest lesson learned you’ve personally learned about the use of AI and HCI in society?
- What is the most critical area that has been neglected in the research on AI ethics and HCI?

After the initial question, the moderators will display audience-focused questions. The audience can type their responses into Slido and the responses can be displayed in near-real time. The moderators will go through each question one-at-a-time to invoke panelist reactions, commentary, and to elicit audience questions/comments.

- Where do you see the biggest opportunity for AI-enabled HCI within society?
- What do you think are the biggest threats associated with the use of AI in society?
- Who has the greatest responsibility for Responsible use of AI? (multiple choice option)
 - ☐ The AI designers
 - ☐ The company hosting the AI
 - ☐ The government to regulate AI
 - ☐ The user (not the developer) of the AI
 - ☐ The AI
- What is the best way to promote responsible use of AI? (select all that apply)
 - ☐ AI literacy training
 - ☐ Effective human-centered interfaces
 - ☐ Effective guardrails to monitor the AI and enforce constraints on AI
 - ☐ Data cards that highlight the purpose, strengths, and weaknesses of using AI in HCI?
 - ☐ Create an effective human-AI team
 - ☐ Something else (write it in)
- Should AI be a teammate or a tool? (select one)
- When AI is used in HCI, does that change whether the AI is a teammate or tool?
- What are some potential pitfalls of using HCI to create a technology as a teammate versus as a tool?
- What are some potential benefits of using HCI to create a technology as a teammate versus as a tool?
- What are some tasks that should never be delegated to AI?
- What should we have asked about AI in HCI that has not yet been asked?
- How does RAI change, if at all, when using AI in HCI research?

For each of the questions, the responses from the audience will be displayed in real-time and the panelists will be asked to respond/comment.

The session will also build in a short amount of time at the end for the audience to pose their own questions to the panel.

At the end of the session, a post-session Slido poll will ask audience members once again to select which ethical or RAI principle(s) the research should focus on next to advance HCI research. The intention is to gauge whether the session led to any opinion change.

This configuration puts the audience in the driver's seat for this panel which should prove to be highly interactive, highly informative and based in the SME panelists domain of expertise, and hopefully entertaining.

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